

Abhinav Kumar

Patna, India

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Education

National Institute of Technology, Patna	2023 – 2028
B.Tech + M.Tech in Computer Science and Engineering (Cyber Security)	CGPA: 7.86 / 10.0
Senior Secondary (Class XII), CBSE	2021
Percentage: 90.8%	
Secondary (Class X), CBSE	2019
Percentage: 95%	

Experience

GYB Club, NIT Patna	Remote
<i>Web Development Co-Lead</i>	Mar 2024 – Sep 2025
- Led development of scalable MERN stack web applications with modular frontend and backend architecture. - Designed secure RESTful APIs implementing JWT-based authentication and RBAC for role-specific access. - Implemented backend validation, error handling, and audit logging to track user actions and system events. - Developed reusable React components and optimized MongoDB queries to improve performance and maintainability. - Coordinated with team members using Git-based workflows, reviewing code and ensuring timely feature delivery.	

Projects

AI-Powered Study Platform (Web + GenAI)
<i>MERN Stack, GenAI APIs, JWT, RBAC, MFA</i>
- Built an AI-driven study platform enabling users to upload learning material and receive AI-generated summaries, explanations, and practice questions. - Secured the platform using JWT authentication, RBAC, and Multi-Factor Authentication (MFA) for students and administrators. - Designed audit logs to monitor uploads, AI usage, and assessments, improving transparency and reliability. - Engineered scalable REST APIs and optimized database schemas to support concurrent users.
System Monitoring & Automation Tool (Python)
<i>Python, Scheduling, Logging</i>
- Developed Python automation to monitor system metrics, process logs, and generate scheduled reports. - Implemented structured logging and audit trails with alerting for failures and anomalies.
Secure Smart Parking Android Application
<i>Android, REST APIs, JWT, RBAC</i>
- Built an Android application for real-time parking discovery and booking. - Secured backend APIs using JWT, RBAC, and OTP-based verification.

Technical Skills

Languages: Java, C++, JavaScript, TypeScript, SQL
Frameworks: React, Next.js, Express.js, Tailwind CSS
Cloud & Databases: AWS, MongoDB, MySQL
Core CS: Data Structures & Algorithms, OOP, OS, DBMS

Positions of Responsibility

Web Development Co-Lead — Web & Coding Club, NIT Patna	2024 – Present
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Achievements

- Knight @ LeetCode** (1875), **specialist @ Codeforces** (1402), **3-Star @ CodeChef**
- Selected for **Adobe Hackathon** among **200,000+ participants** (Top 0.5%).
- Solved **1,000+ DSA problems** across LeetCode, Codeforces, AtCoder, CSES, and contests.